

Cover Page
SQL DATA ANALYSIS
Project 3
Industrial Training Kit
Submitted By
Harshath Mohammad
Submitted To
DecodeLabs
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Introduction

Data Visualization is the process of representing data graphically to make patterns, trends, and relationships easier to understand. In this project, an e-commerce sales dataset was analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn. Multiple charts were created to effectively communicate business insights.

Objectives

- Learn data visualization techniques.
- Create meaningful charts.
- Analyze sales data visually.
- Identify business trends.
- Present data effectively.

Tools Used

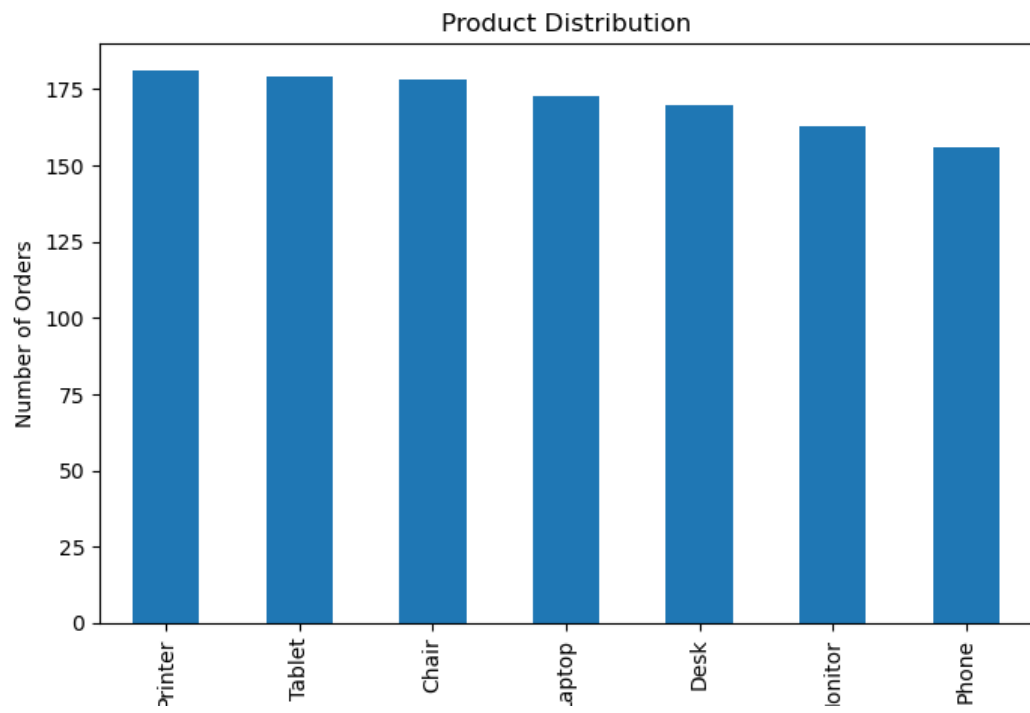
Tool	Purpose
Python	Programming
Pandas	Data manipulation
Matplotlib	Chart creation
Seaborn	Statistical visualization
Jupyter Notebook	Development environment

Dataset Description

The dataset contains **1200 e-commerce sales records** with **14 columns**, including order details, products, payment methods, quantities, and total sales values.

Data Visualization

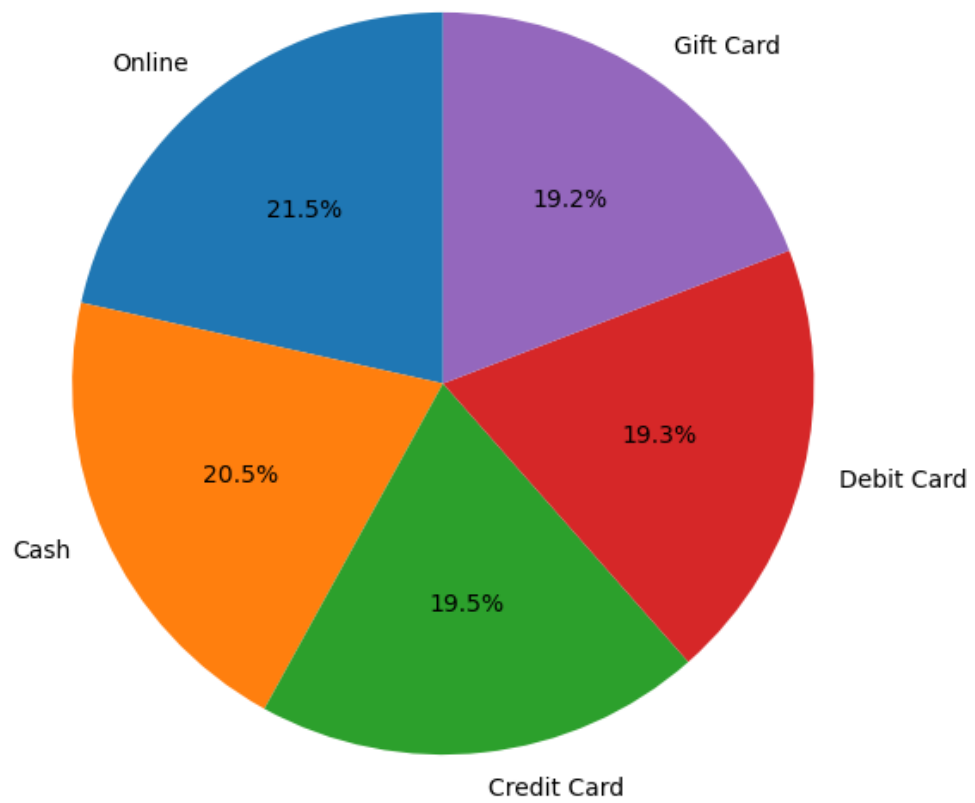
Bar Chart



Displays the number of orders for each product category.

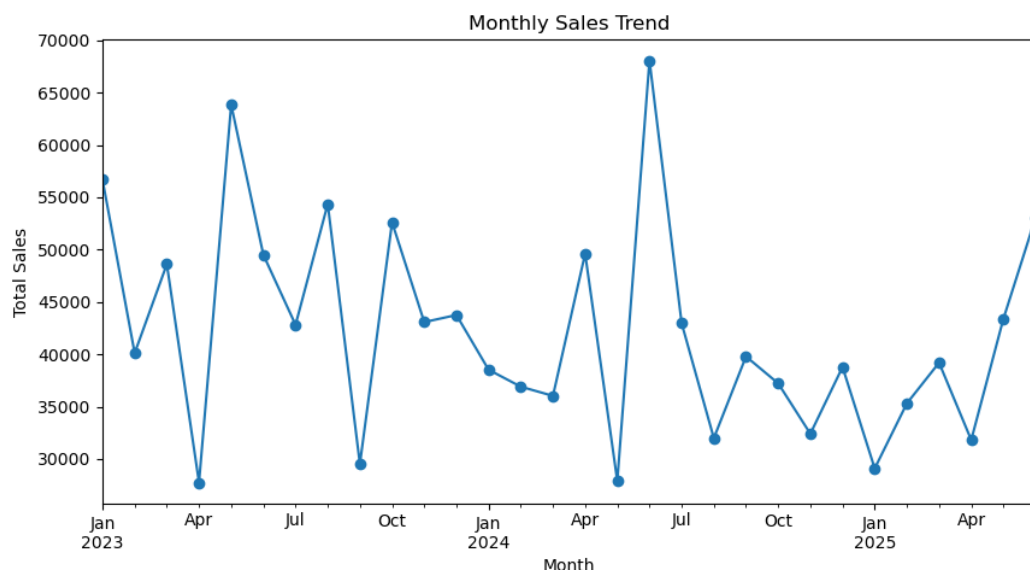
Pie Chart

Payment Method Distribution



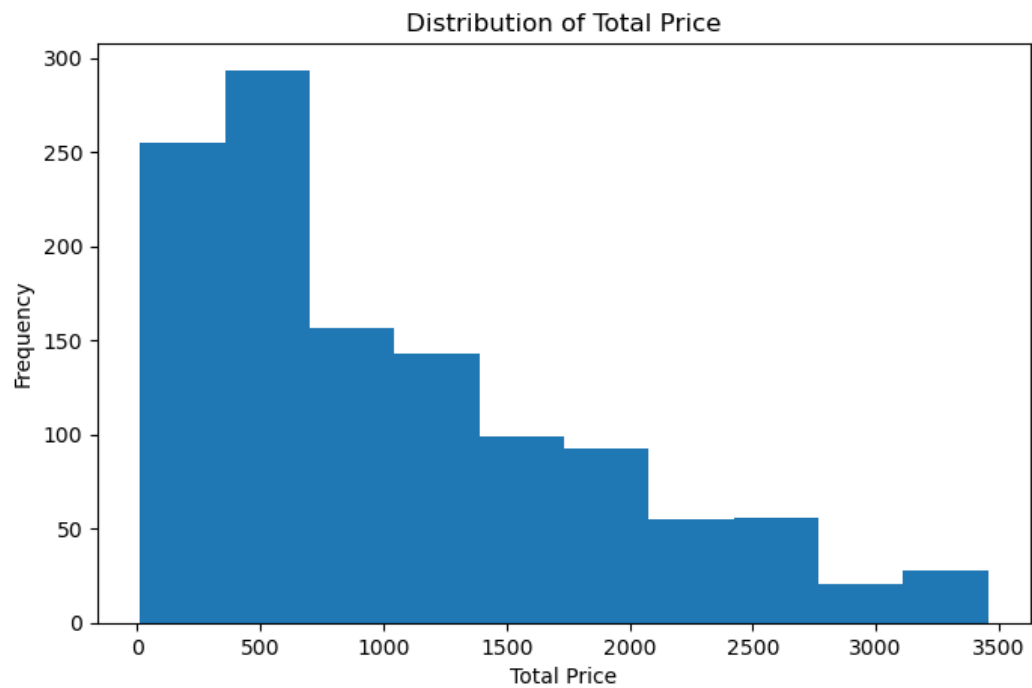
Shows the percentage distribution of payment methods.

Line Chart



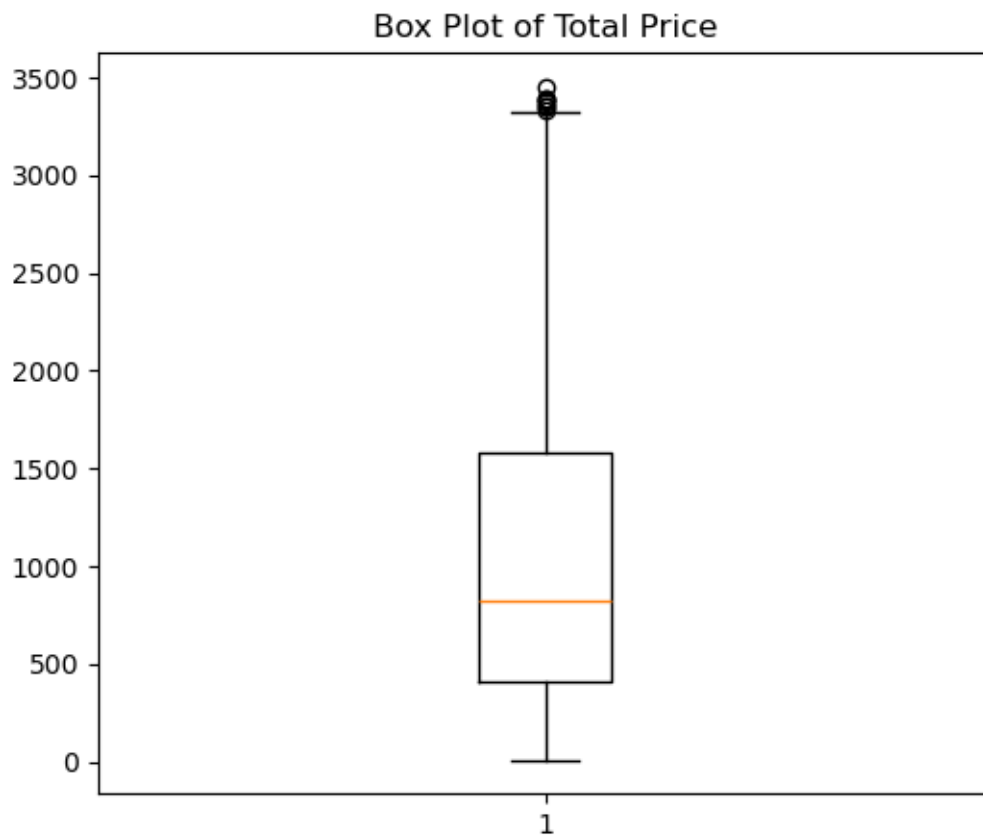
Illustrates the monthly sales trend over time.

Histogram



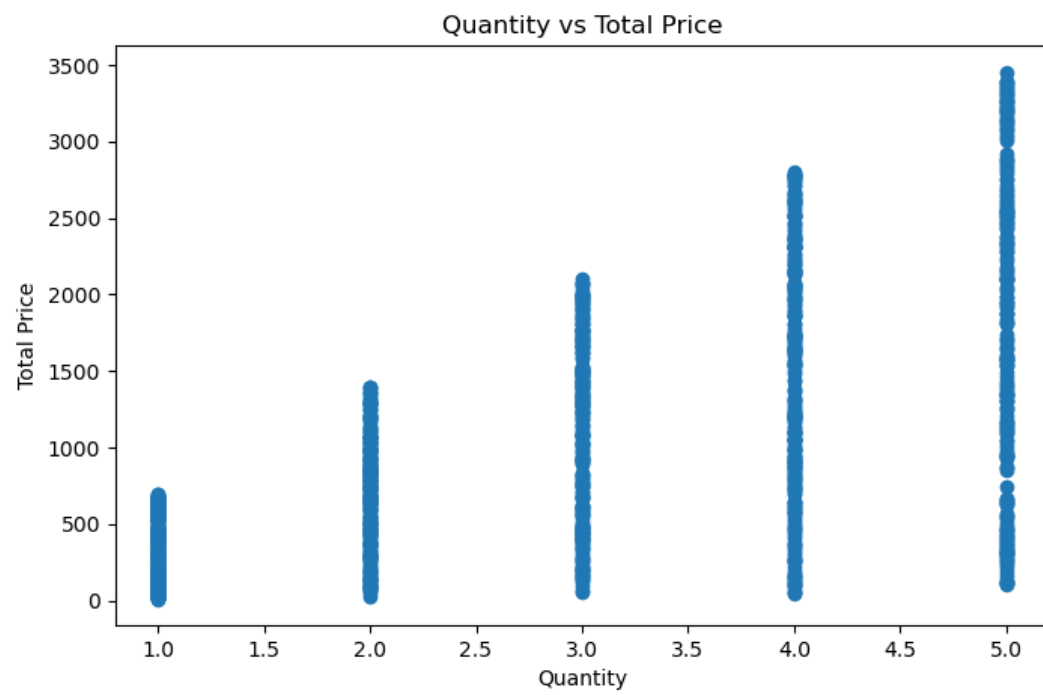
Shows the distribution of total order values.

Box Plot



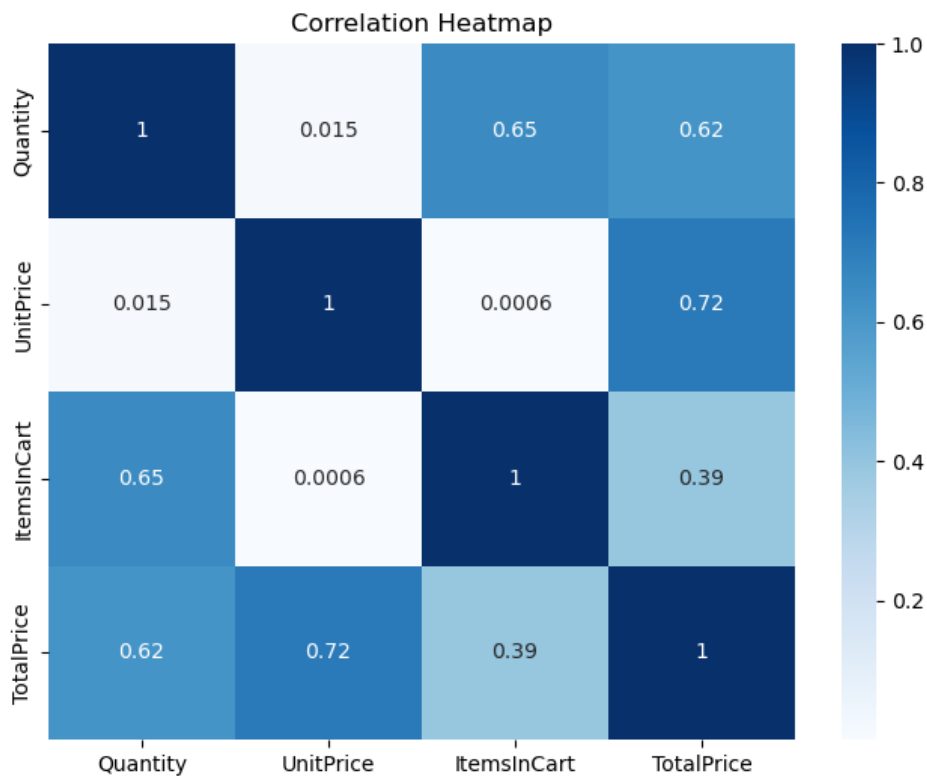
Highlights the spread of total prices and potential outliers.

Scatter Plot



Shows the relationship between quantity purchased and total price.

Heatmap



Displays correlations between numerical variables.

Key Insights

- Product demand varies across categories.
- Customers use multiple payment methods with fairly balanced usage.
- Sales fluctuate over different months.
- Most orders fall within a common price range.
- Higher quantities generally lead to higher total prices.
- Numerical features show useful correlations for analysis

Conclusion

This project demonstrated how data visualization helps transform raw data into meaningful insights. By creating different chart types, patterns in sales, customer behavior, and pricing became easier to understand. These visualizations support better business decision-making and improve data interpretation.